

BIERE, Switzerland

WMO: 067040

Lat: 46.5250N

Lon: 6.3425E

Elev: 2244

StdP: 13.54

Time Zone: 1.00 (EUC)

Period: 10-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	16.1	20.3	8.9	9.4	20.1	14.0	12.2	23.6	24.2	33.8	22.4	27.9	8.9	40	0.510

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.0	84.5	67.3	81.2	65.9	78.0	64.4	68.8	81.0	67.2	78.8	65.7	76.1	6.4	130

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
64.7	99.5	74.5	63.2	94.1	72.4	61.8	89.6	70.7	34.4	81.5	33.0	78.7	31.8	76.3	77.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 19.9	16.9	14.5	DB	13.6	90.1	3.6	3.9	11.0	92.9	8.9	95.2	6.8	97.4	4.2	100.2	(4)
(5)			WB	12.1	72.3	4.0	2.3	9.2	73.9	6.9	75.3	4.7	76.6	1.8	78.2	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	49.4	34.1	34.1	42.6	49.3	54.0	62.2	66.0	64.6	58.2	50.5	41.2	35.2		
	DBStd	13.04	6.48	7.81	6.84	6.59	6.36	6.33	5.95	5.82	5.31	6.20	6.53	6.34		
	HDD50	2110	492	446	240	92	31	1	0	0	4	72	272	460		
	HDD65	5890	957	865	693	470	343	126	59	79	211	448	714	925		
	CDD50	1901	0	1	12	71	155	367	495	454	250	88	8	0		
	CDD65	208	0	0	0	0	2	42	89	68	7	0	0	0		
	CDH74	1788	0	0	0	2	28	374	782	542	60	0	0	0		
	CDH80	416	0	0	0	0	0	83	217	115	1	0	0	0		
	WSAvg	5.7	6.3	6.7	7.0	6.2	5.9	5.3	5.1	4.6	5.1	5.1	5.6	5.9		
	PrecAvg	50.1	4.1	3.9	3.4	3.6	4.2	4.2	4.3	4.2	4.1	4.5	4.8	4.8		
(15) Precipitation	PrecMax	60.7	10.6	12.2	13.9	6.3	7.4	8.5	8.6	9.7	9.9	8.7	12.6	11.4		
	PrecMin	37.2	0.8	0.6	0.8	0.8	1.3	1.7	1.9	0.8	1.5	1.0	0.6	0.1		
	PrecStd	6.7	2.4	2.6	2.8	1.8	1.7	1.8	1.6	1.9	2.3	2.2	2.8	2.6		
	PrecStd	6.7	2.4	2.6	2.8	1.8	1.7	1.8	1.6	1.9	2.3	2.2	2.8	2.6		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	52.6	56.9	63.9	73.1	77.4	86.2	90.6	86.3	78.9	69.6	60.0	53.2		
		MCWB	45.6	45.8	49.3	55.4	60.6	70.3	66.6	67.3	62.7	58.1	52.0	46.0		
	2%	DB	48.7	51.3	60.8	68.7	73.6	82.3	85.4	83.2	75.5	66.5	56.9	49.5		
		MCWB	44.8	43.4	48.4	53.3	59.5	67.4	66.5	67.2	62.7	56.7	50.9	45.6		
	5%	DB	46.4	47.8	57.5	65.4	70.3	78.2	81.7	80.0	71.9	63.1	53.9	47.4		
		MCWB	43.4	41.6	46.9	52.1	57.3	65.1	65.4	66.3	61.0	56.3	49.7	44.3		
	10%	DB	43.5	44.7	53.6	61.6	66.6	74.7	78.0	76.5	68.8	60.0	51.2	45.2		
		MCWB	41.3	40.5	45.1	50.6	55.8	63.1	64.0	64.7	59.9	54.8	48.1	42.3		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	48.1	47.3	51.3	57.5	63.3	72.8	69.7	70.2	65.7	60.8	54.1	48.5		
		MCDB	50.2	52.0	60.4	70.0	73.7	84.7	83.5	81.6	72.6	65.5	56.9	50.6		
	2%	WB	45.7	44.9	49.5	54.9	60.9	68.6	68.0	68.6	63.9	58.7	52.1	46.5		
		MCDB	47.8	50.6	57.5	66.3	71.3	79.6	81.9	80.3	72.6	63.2	55.2	48.8		
	5%	WB	43.5	42.7	47.8	53.0	58.7	66.4	66.6	67.0	62.4	57.2	50.4	44.9		
		MCDB	46.0	46.9	55.8	63.1	67.0	76.6	79.2	77.8	70.5	61.7	53.6	47.4		
	10%	WB	41.6	40.5	46.0	51.1	56.7	64.0	65.2	65.5	60.7	55.5	48.4	42.7		
		MCDB	43.8	44.4	52.9	60.2	64.9	73.2	76.2	75.0	67.5	59.4	50.5	45.2		
(35) Mean Daily Temperature Range	5% DB	MDBR	9.7	12.0	15.0	17.4	17.0	18.0	19.0	18.8	17.1	14.2	10.9	10.9		
		MCDBR	12.1	18.5	21.5	23.4	22.8	23.3	25.1	23.3	21.8	18.7	14.6	13.5		
		MCWBR	8.9	12.1	11.9	12.0	11.6	12.1	10.7	10.6	11.3	11.3	10.2	10.1		
		MCDBR	11.1	16.2	19.2	21.7	20.8	22.2	22.6	22.1	19.5	16.8	13.9	12.4		
	5% WB	MCWBR	8.8	11.6	11.4	11.8	11.6	12.4	10.6	10.7	10.7	10.8	10.6	10.0		
(40) Clear-Sky Solar Irradiance	taub		0.283	0.306	0.341	0.376	0.385	0.386	0.384	0.376	0.357	0.340	0.308	0.282		
	taud		2.457	2.410	2.381	2.309	2.326	2.359	2.356	2.389	2.426	2.466	2.501	2.475		
	Ebn at Noon		261	276	280	279	279	278	276	272	259	249	249	249		
	Edh at Noon		21	27	32	38	39	38	38	35	31	26	20	19		
(44) All-Sky Solar Radiation	RadAvg		424	700	1064	1392	1593	1889	1853	1613	1263	813	453	345		
	RadStd		49	101	150	197	142	177	159	144	108	82	58	54		

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (90 neighbors)	N/A	N/A	N/A	N/A	N/A	+0.67	+0.49	N/A	N/A	N/A	N/A

Nomenclature: See separate page